Database System Project

Phase 1 + phase 2 + phase 3 + phase 4 + bonus

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Bonus:

Code: https://github.com/saadmajaly/bank_management

Video: https://www.youtube.com/watch?v=kS_RWYVid3s

Bank Accounts Management System

Our main Idea is to make a database system the manages accounts and handle the data of the bank account and the other things related to the accounts.

Our system requirements:

-Each **customer** has a first name(FName), middle name(MName), last name(LName), the customer's nationality(Nationality), also it should have the customer's approved personal document number(PDNum), the customer's approved residence proof document number(ResProofDoc), and the customer's phone number(PhoneNum).

-Each **bank account** has automatically generated account number that subject to the International Format for Account Numbers(AccountNum), also the money balance in the account(Balance), every bank account has to be one of the following bank account types:

-**Saving Account**: the saving account has one or more currencies(Cur) in which the customer can deposit his money (JOD, EUR, USD, GBP), also the number of the ATM deposit card(CardNum), also it has calculated profit percentage(profit) and the profit rate is 0.375% of the smallest amount deposited in the account every 3 months.

-**Current Account**: the current account has one or more currencies(Cur) in which the customer can deposit his money (JOD, EUR, USD, GBP), and it has the number of the deposit card(DCardNum) or the number of the credit card(CCardNum) or both of them, also it has The bank check book number(Check BookNum) and the number of bank checks in it.

-**Investments Deposits Account**: The Investments Deposit Account has one currency(Cur) in which the customer can deposit his money (JOD or USD), also it has Deposit period(DPeriod) in days that when this period ends the profits will be given to the customer also it should have the profit(Profits) that is calculated using this formula ((the balance × profit rate × no. of days that the balance will be in the account)/365).

-**Corporate Account**: A Corporate Account has the corporate name(CorName) and the corporate registration number(CorRegNumber) in the Ministry of Industry and Trade.

-**Loan**: the data that the loan should have is the account that want to take the loan information(AccountNum), and it should calculate the maximum amount of the loan(MaxLoan) that can be given to the account and if the account can afford that loan based on the income average and the previous loans and the current loans that the customer is paying back to the bank, also it should have the loan amount(LoanAmount) and the monthly amount(MonAmount) that have to be taken from the account and the number of times that the customer delayed the monthly payment(Delays).

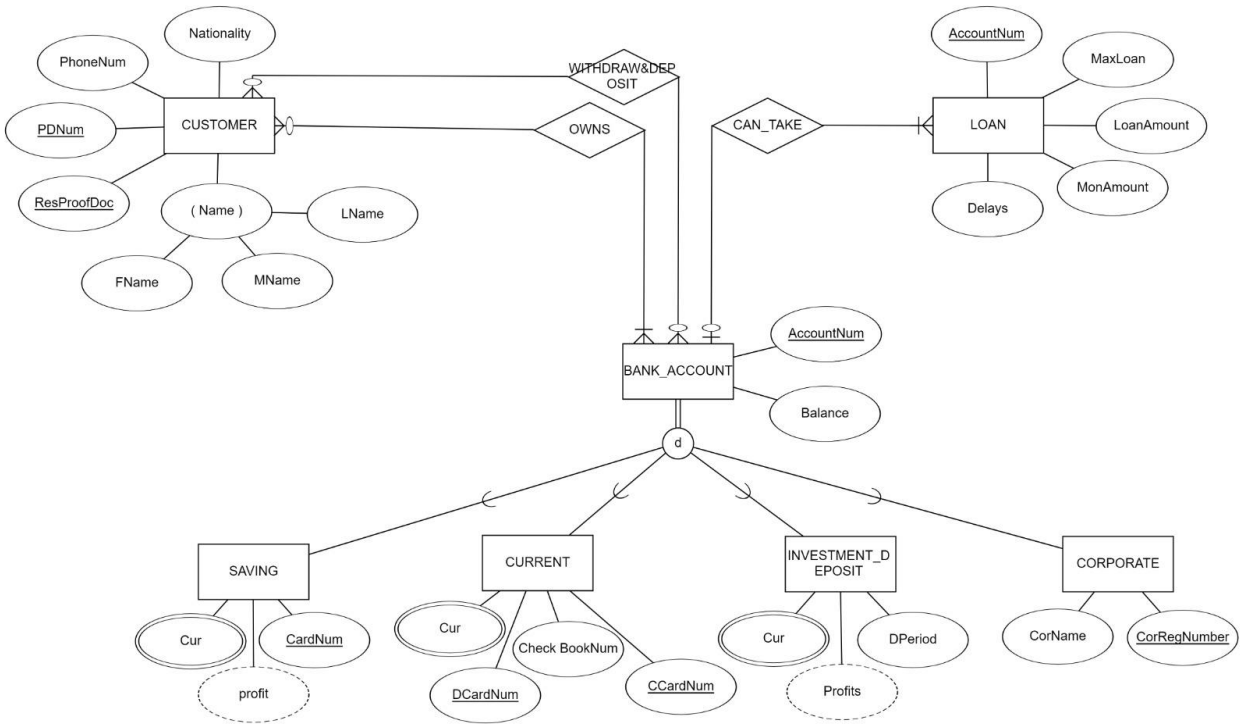
-every customer can have more than one account.

- every account can take many types of loans from the bank and more than one loan.

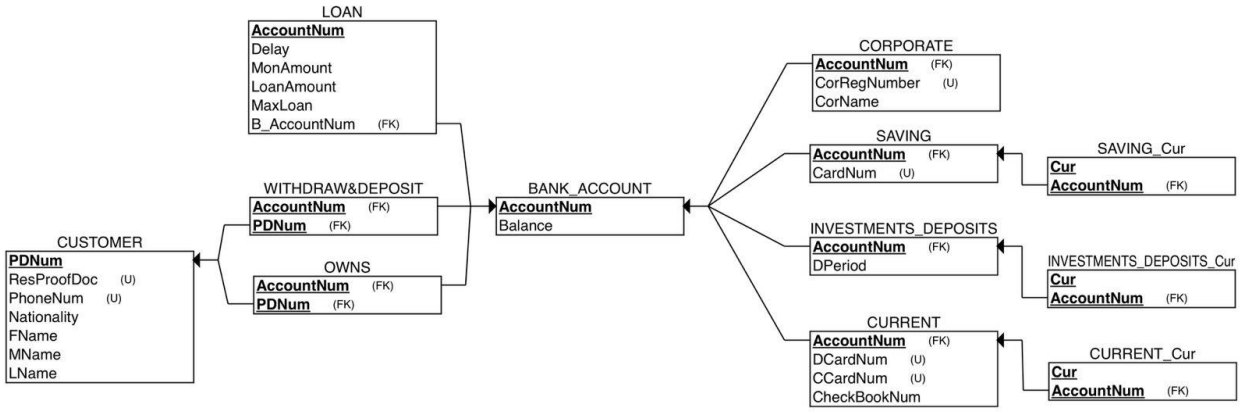
-the Corporate account can be for more than one customer who is a shareholder in the company.

-every customer can withdraw and deposit money from any of his accounts.

ER Diagram:



Schema Diagram:



SQL:

drop table CUSTOMER cascade constraints;

drop table BANK_ACCOUNT cascade constraints;

drop table LOAN cascade constraints;

drop table WITHDRAW_DEPOSIT cascade constraints;

drop table OWNS cascade constraints;

drop table CORPORATE_ACC cascade constraints;

drop table SAVING_ACC cascade constraints;

drop table INV_DEP_ACC cascade constraints;

drop table CURRENT_ACC cascade constraints;

drop table SAVING_CUR cascade constraints;

drop table INV_DEP_CUR cascade constraints;

drop table CURRENT_CUR cascade constraints;

```
SQL Plus

SQL> CREATE TABLE CUSTOMER (PDNUM NUMBER (11) CONSTRAINT PDNUM_PK PRIMARY KEY, RESPROOFDOC NUMBER (12) CONSTRAINT RESPROOFDOC_UQ UNIQUE, PHONENUM NUMBER (10
) CONSTRAINT PHONENUM_UQ UNIQUE, NATIONALITY VARCHAR2(20), FNAME VARCHAR2(15) CONSTRAINT FNAME_NN NOT NULL, MNAME VARCHAR2(15) CONSTRAINT MNAME_NN NOT NULL,
 LNAME VARCHAR2(20) CONSTRAINT LNAME_NN NOT NULL);

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE BANK_ACCOUNT (ACCOUNTNUM NUMBER(16) CONSTRAINT ACCOUNTNUM_PK PRIMARY KEY, BALANCE NUMBER(10) CONSTRAINT BALANCE_NN NOT NULL);

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE LOAN (ACCOUNTNUM NUMBER(16) CONSTRAINT ACCOUNTNUM_LOAN_PK PRIMARY KEY, DELAY NUMBER(3), MONAMOUNT NUMBER(10) CONSTRAINT MONAMOUNT_C CHECK
(MONAMOUNT > -1), LOANAMOUNT NUMBER(10) CONSTRAINT LOANAMOUNT_C CHECK (LOANAMOUNT > -1), MAXAMOUNT NUMBER(10) CONSTRAINT MAXAMOUNT_NN NOT NULL, B_ACCOUNTNUM
 NUMBER(16) CONSTRAINT B_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE);

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE WITHDRAW_DEPOSIT (ACCNUM NUMBER(16) CONSTRAINT WD_ACCNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE, PDNUM NUMBER(11) CONSTR
AINT WD_PDNUM_FK REFERENCES CUSTOMER(PDNUM) ON DELETE CASCADE, CONSTRAINT WD_ACC_PD_PK PRIMARY KEY(ACCNUM, PDNUM));

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE OWNS (ACCNUM NUMBER(16) CONSTRAINT O_ACCNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE, PDNUM NUMBER(11) CONSTRAINT O_PDNUM_
FK REFERENCES CUSTOMER(PDNUM) ON DELETE CASCADE, CONSTRAINT O_ACC_PD_PK PRIMARY KEY(ACCNUM, PDNUM));

Table created.

SQL>
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SQL Plus
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE CORPORATE_ACC (ACCNUM NUMBER(16) CONSTRAINT C_ACCNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT C_ACCNUM_PK PRIMA
RY KEY, CORREGNUMBER NUMBER(21) CONSTRAINT CORREGNUM_UQ UNIQUE, CORNAME VARCHAR2(20) CONSTRAINT CORNAME_NN NOT NULL);

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE SAVING_ACC (ACCNUM NUMBER(16) CONSTRAINT S_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT S_ACCOUNTNUM_PK
PRIMARY KEY , CARDNUM NUMBER(16) CONSTRAINT CARDNUM_UQ UNIQUE);

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE INV_DEP_ACC (ACCNUM NUMBER(16) CONSTRAINT IND_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT IND_ACCOUNTNU
M_PK PRIMARY KEY , DPERIOD NUMBER(3) CONSTRAINT DPERIOD_NN NOT NULL);

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE CURRENT_ACC (ACCNUM NUMBER(16) CONSTRAINT CR_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT CR_ACCOUNTNUM_
PK PRIMARY KEY , DCARDNUM NUMBER(16) CONSTRAINT DCARDNUM_UQ UNIQUE, CCARDNUM NUMBER(16) CONSTRAINT CCARDNUM_UQ UNIQUE, CHECKBOOKNUM NUMBER(6));

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE SAVING_CUR (CUR VARCHAR2(3), ACCNUM NUMBER(16) CONSTRAINT SC_ACCNUM_FK REFERENCES SAVING_ACC(ACCNUM) ON DELETE CASCADE, CONSTRAINT SC_CUR_
ACCNUM_PK PRIMARY KEY(CUR,ACCNUM));

Table created.

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SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE INV_DEP_CUR (CUR VARCHAR2(3), ACCNUM NUMBER(16) CONSTRAINT IND_ACCNUM_FK REFERENCES INV_DEP_ACC(ACCNUM) ON DELETE CASCADE, CONSTRAINT INV_
CUR_ACCNUM_PK PRIMARY KEY(CUR,ACCNUM));

Table created.

SQL>
SQL> ------------------------------------------------------------------------------
SQL>
SQL> CREATE TABLE CURRENT_CUR (CUR VARCHAR2(3), ACCNUM NUMBER(16) CONSTRAINT CC_ACCNUM_FK REFERENCES CURRENT_ACC(ACCNUM) ON DELETE CASCADE, CONSTRAINT CC_CU
R_ACCNUM_PK PRIMARY KEY(CUR,ACCNUM));

Table created.
```

CREATE TABLE CUSTOMER (PDNUM NUMBER (11) CONSTRAINT PDNUM_PK PRIMARY KEY, RESPROOFDOC NUMBER (12) CONSTRAINT RESPROOFDOC_UQ UNIQUE, PHONENUM NUMBER (10) CONSTRAINT PHONENUM_UQ UNIQUE, NATIONALITY VARCHAR2(20), FNAME VARCHAR2(15) CONSTRAINT FNAME_NN NOT NULL, MNAME VARCHAR2(15) CONSTRAINT MNAME_NN NOT NULL, LNAME VARCHAR2(20) CONSTRAINT LNAME_NN NOT NULL);

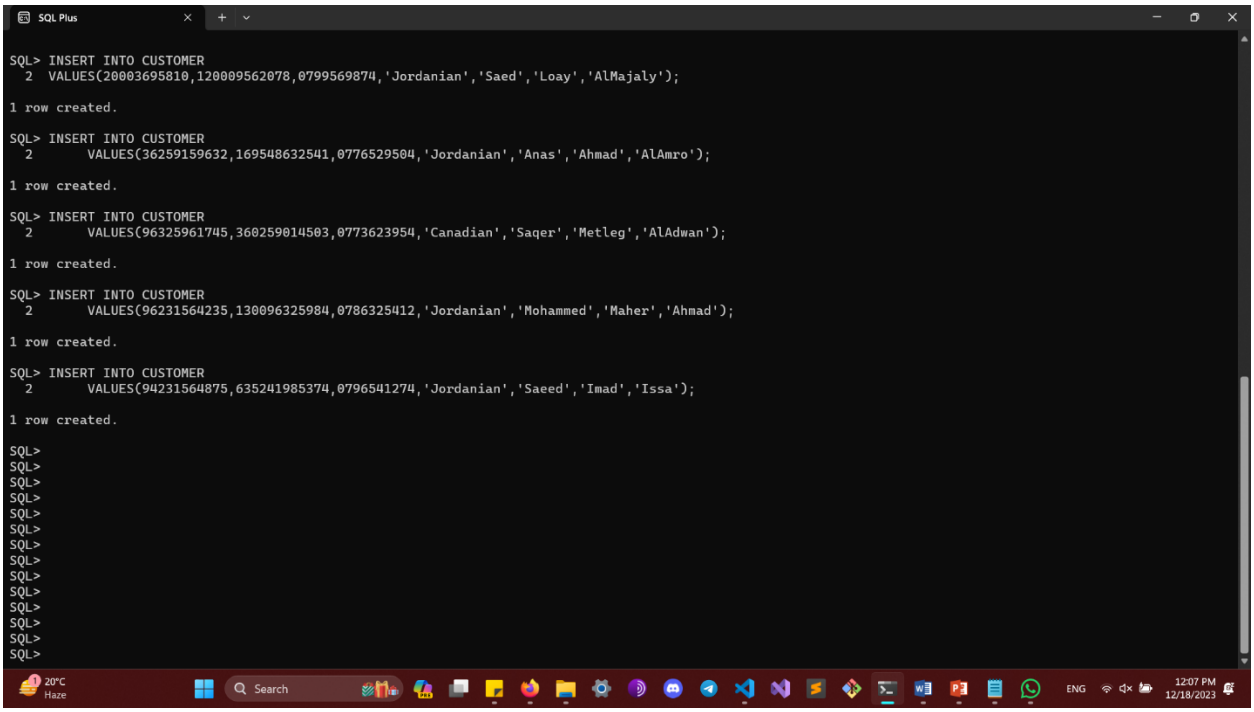
INSERT INTO CUSTOMER
VALUES(20003695810,120009562078,0799569874,'Jordanian','Saed','Loay','AlMajaly');

INSERT INTO CUSTOMER
VALUES(36259159632,169548632541,0776529504,'Jordanian','Anas','Ahmad','AlAmro');

INSERT INTO CUSTOMER
VALUES(96325961745,360259014503,0773623954,'Canadian','Saqer','Metleg','AlAdwan');

INSERT INTO CUSTOMER
VALUES(96231564235,130096325984,0786325412,'Jordanian','Mohammed','Maher','Ahmad');

INSERT INTO CUSTOMER
VALUES(94231564875,635241985374,0796541274,'Jordanian','Saeed','Imad','Issa');



CREATE TABLE BANK_ACCOUNT (ACCOUNTNUM NUMBER(16) CONSTRAINT ACCOUNTNUM_PK PRIMARY KEY, BALANCE NUMBER(10) CONSTRAINT BALANCE_NN NOT NULL);

INSERT INTO BANK_ACCOUNT VALUES(4699632545639255,200);

INSERT INTO BANK_ACCOUNT VALUES(6321564236159632,53966.69);

INSERT INTO BANK_ACCOUNT VALUES(6321564236159966,153325.91);

```
INSERT INTO BANK_ACCOUNT VALUES(6321564236783215,2678.45);

INSERT INTO BANK_ACCOUNT VALUES(6321564239635421,15633.23);

INSERT INTO BANK_ACCOUNT VALUES(1234567890123456,567890.12);

INSERT INTO BANK_ACCOUNT VALUES(9876543210987654,190123.45);

INSERT INTO BANK_ACCOUNT VALUES(3141592653589793,65432.10);

INSERT INTO BANK_ACCOUNT VALUES(2718281828459045,456789.12);

INSERT INTO BANK_ACCOUNT VALUES(1618033988749894,189012.34);

INSERT INTO BANK_ACCOUNT VALUES(1414213562373095,76543.21);

INSERT INTO BANK_ACCOUNT VALUES(1732050807568877,345678.91);

INSERT INTO BANK_ACCOUNT VALUES(2236067977499789,78901.23);

INSERT INTO BANK_ACCOUNT VALUES(2449489742783178,87654.32);

INSERT INTO BANK_ACCOUNT VALUES(3162277660168379,234567.89);

INSERT INTO BANK_ACCOUNT VALUES(3872983346207416,156789.01);

INSERT INTO BANK_ACCOUNT VALUES(4242640687119285,98765.43);

INSERT INTO BANK_ACCOUNT VALUES(4582575694955840,123456.78);
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INSERT INTO BANK_ACCOUNT VALUES(5477225575051661,34567.89);

INSERT INTO BANK_ACCOUNT VALUES(6324555320336758,54321.09);

```
SQL> INSERT INTO BANK_ACCOUNT
  2  VALUES(4699632545639255,200);

1 row created.

SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(6321564236159632,53966.69);

1 row created.

SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(6321564236159966,153325.91);

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SQL> INSERT INTO BANK_ACCOUNT
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SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(6321564239635421,15633.23);

1 row created.

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SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(1234567890123456,567890.12);

1 row created.

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SQL> ----------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(9876543210987654,190123.45);

1 row created.

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SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(3141592653589793,65432.10);

1 row created.

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1 row created.

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SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(3872983346207416,156789.01);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(4242640687119285,98765.43);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
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SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(4582575694955840,123456.78);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(5477225575051661,34567.89);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO BANK_ACCOUNT
  2       VALUES(6324555320336758,54321.09);

1 row created.

SQL>
SQL> ----------------------------------------------------------------------------------------------
SQL>
```

CREATE TABLE LOAN (ACCOUNTNUM NUMBER(16) CONSTRAINT ACCOUNTNUM_LOAN_PK PRIMARY KEY, DELAY NUMBER(3), MONAMOUNT NUMBER(10) CONSTRAINT MONAMOUNT_C CHECK (MONAMOUNT > -1), LOANAMOUNT NUMBER(10) CONSTRAINT LOANAMOUNT_C CHECK

(LOANAMOUNT > -1), MAXAMOUNT NUMBER(10) CONSTRAINT MAXAMOUNT_NN NOT NULL, B_ACCOUNTNUM NUMBER(16) CONSTRAINT B_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE);

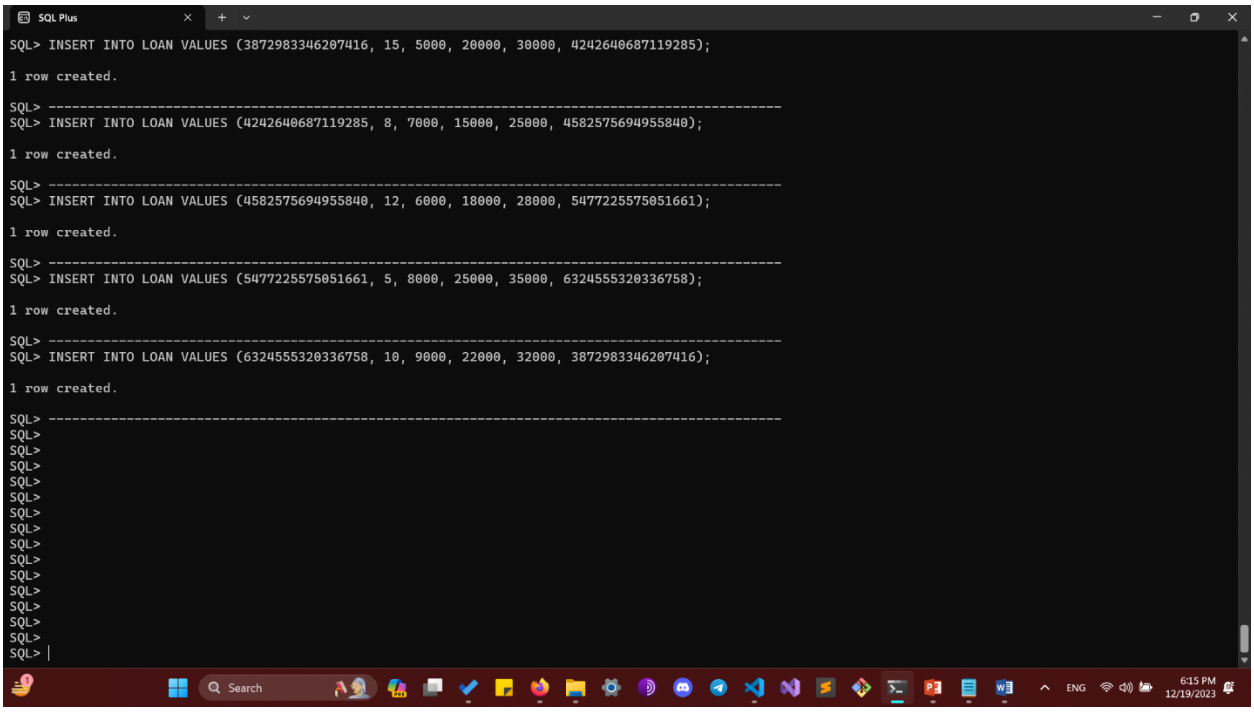
INSERT INTO LOAN VALUES (3872983346207416, 15, 5000, 20000, 30000, 3872983346207416);

INSERT INTO LOAN VALUES (4242640687119285, 8, 7000, 15000, 25000, 4242640687119285);

INSERT INTO LOAN VALUES (4582575694955840, 12, 6000, 18000, 28000, 4582575694955840);

INSERT INTO LOAN VALUES (5477225575051661, 5, 8000, 25000, 35000, 5477225575051661);

INSERT INTO LOAN VALUES (6324555320336758, 10, 9000, 22000, 32000, 6324555320336758);



CREATE TABLE WITHDRAW_DEPOSIT (ACCNUM NUMBER(16) CONSTRAINT WD_ACCNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE, PDNUM NUMBER(11) CONSTRAINT WD_PDNUM_FK REFERENCES CUSTOMER(PDNUM) ON DELETE CASCADE, CONSTRAINT WD_ACC_PD_PK PRIMARY KEY(ACCNUM, PDNUM));

INSERT INTO WITHDRAW_DEPOSIT VALUES (4242640687119285,20003695810);

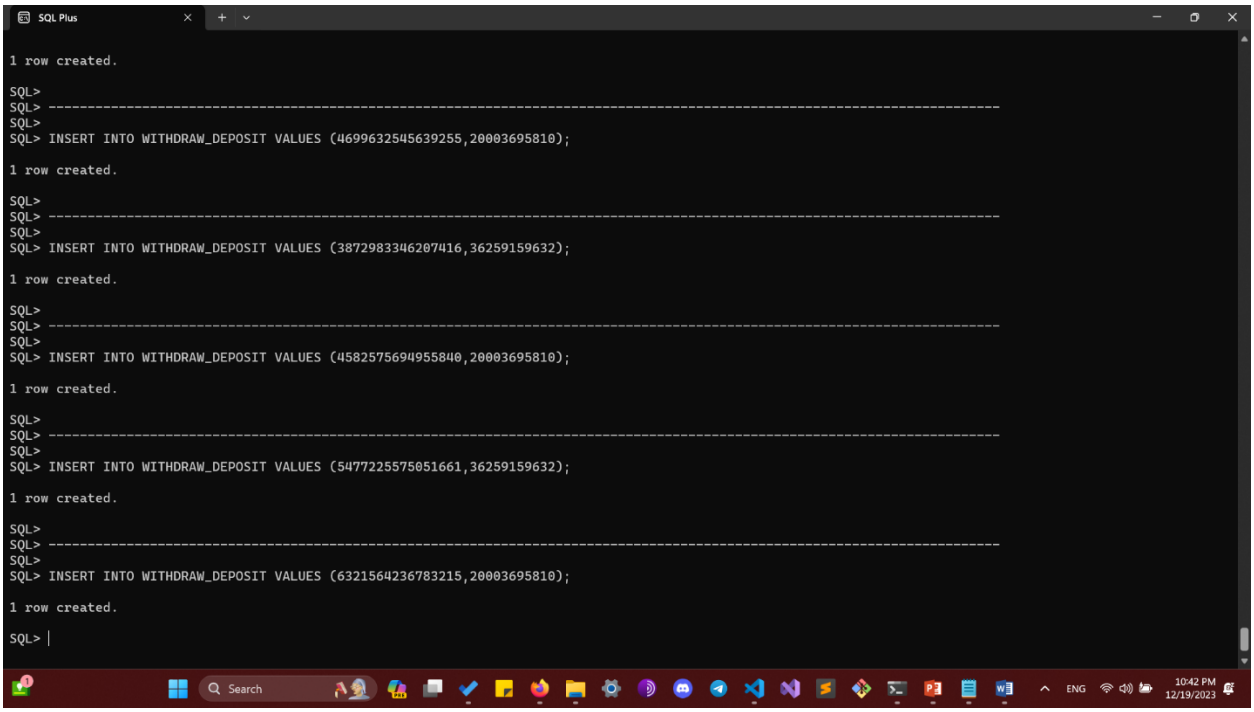
INSERT INTO WITHDRAW_DEPOSIT VALUES (4699632545639255,20003695810);

INSERT INTO WITHDRAW_DEPOSIT VALUES (3872983346207416,36259159632);

INSERT INTO WITHDRAW_DEPOSIT VALUES (4582575694955840,20003695810);

INSERT INTO WITHDRAW_DEPOSIT VALUES (5477225575051661,36259159632);

INSERT INTO WITHDRAW_DEPOSIT VALUES (6321564236783215,20003695810);



CREATE TABLE OWNS (ACCNUM NUMBER(16) CONSTRAINT O_ACCNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE, PDNUM NUMBER(11) CONSTRAINT O_PDNUM_FK REFERENCES CUSTOMER(PDNUM) ON DELETE CASCADE, CONSTRAINT O_ACC_PD_PK PRIMARY KEY(ACCNUM, PDNUM));

INSERT INTO OWNS VALUES (4699632545639255, 20003695810);

INSERT INTO OWNS VALUES (6321564236159632, 20003695810);

INSERT INTO OWNS VALUES (6321564236159966, 20003695810);

INSERT INTO OWNS VALUES (6321564236783215, 20003695810);

INSERT INTO OWNS VALUES (6321564236159632, 36259159632);

INSERT INTO OWNS VALUES (6321564236159966, 36259159632);

INSERT INTO OWNS VALUES (6321564236783215, 36259159632);

INSERT INTO OWNS VALUES (6321564239635421, 36259159632);

INSERT INTO OWNS VALUES (6321564236159632, 96325961745);

INSERT INTO OWNS VALUES (6321564236159966, 96325961745);

INSERT INTO OWNS VALUES (6321564236783215, 96325961745);

INSERT INTO OWNS VALUES (6321564239635421, 96325961745);

INSERT INTO OWNS VALUES (6321564236159632, 96231564235);

INSERT INTO OWNS VALUES (6321564236159966, 96231564235);

INSERT INTO OWNS VALUES (6321564236783215, 96231564235);

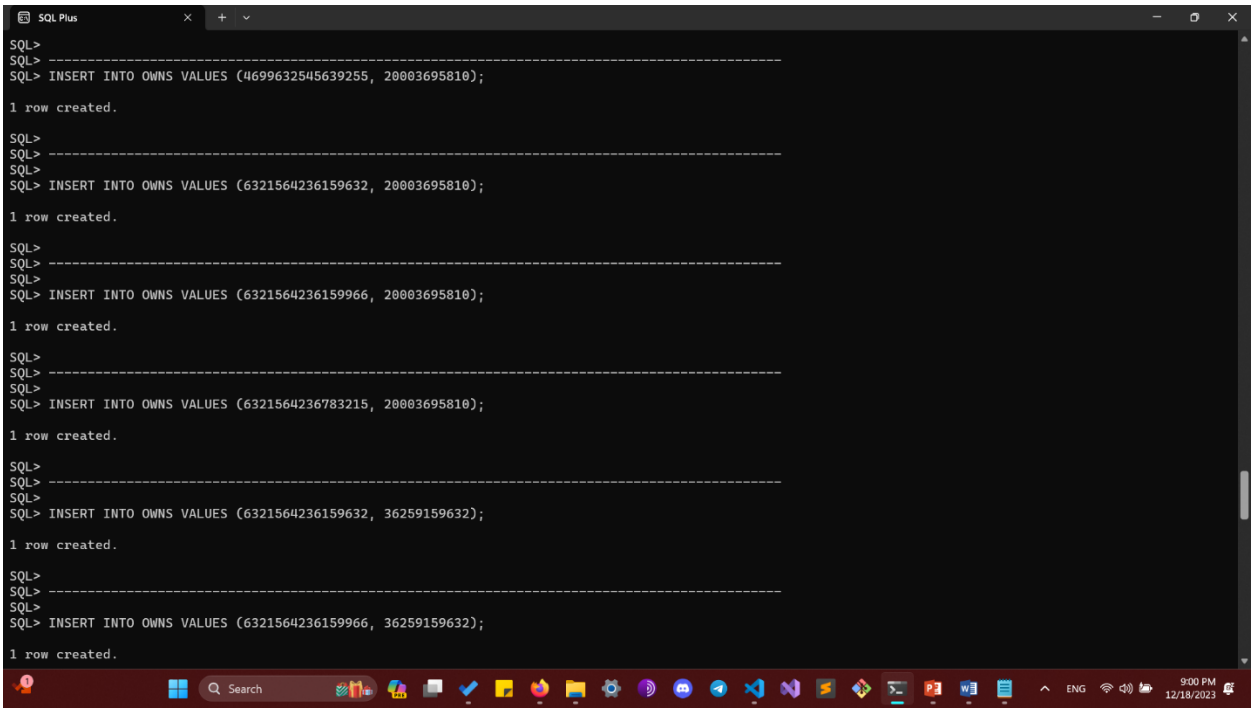
INSERT INTO OWNS VALUES (6321564239635421, 96231564235);

INSERT INTO OWNS VALUES (6321564236159632, 94231564875);

INSERT INTO OWNS VALUES (6321564236159966, 94231564875);

INSERT INTO OWNS VALUES (6321564236783215, 94231564875);

INSERT INTO OWNS VALUES (6321564239635421, 94231564875);



```
SQL> INSERT INTO OWNS VALUES (6321564236783215, 36259159632);

1 row created.

SQL>
SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564239635421, 36259159632);

1 row created.

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SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159632, 96325961745);

1 row created.

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SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159966, 96325961745);

1 row created.

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SQL>
SQL> INSERT INTO OWNS VALUES (6321564236783215, 96325961745);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564239635421, 96325961745);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159632, 96231564235);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159966, 96231564235);

1 row created.

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SQL> INSERT INTO OWNS VALUES (6321564236783215, 96231564235);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564239635421, 96231564235);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159632, 94231564875);

1 row created.

SQL>
SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159966, 94231564875);

1 row created.
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SQL> INSERT INTO OWNS VALUES (6321564236783215, 96231564235);

1 row created.

SQL>
SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564239635421, 96231564235);

1 row created.

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SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159632, 94231564875);

1 row created.

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SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564236159966, 94231564875);

1 row created.

SQL>
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SQL>
SQL> INSERT INTO OWNS VALUES (6321564236783215, 94231564875);

1 row created.

SQL>
SQL> ----------------------------------------------------------------------------------------------
SQL>
SQL> INSERT INTO OWNS VALUES (6321564239635421, 94231564875);

1 row created.

SQL>
```

CREATE TABLE CORPORATE_ACC (ACCNUM NUMBER(16) CONSTRAINT C_ACCNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT C_ACCNUM_PK PRIMARY KEY,

CORREGNUMBER NUMBER(21) CONSTRAINT CORREGNUM_UQ UNIQUE, CORNAME VARCHAR2(20) CONSTRAINT CORNAME_NN NOT NULL);

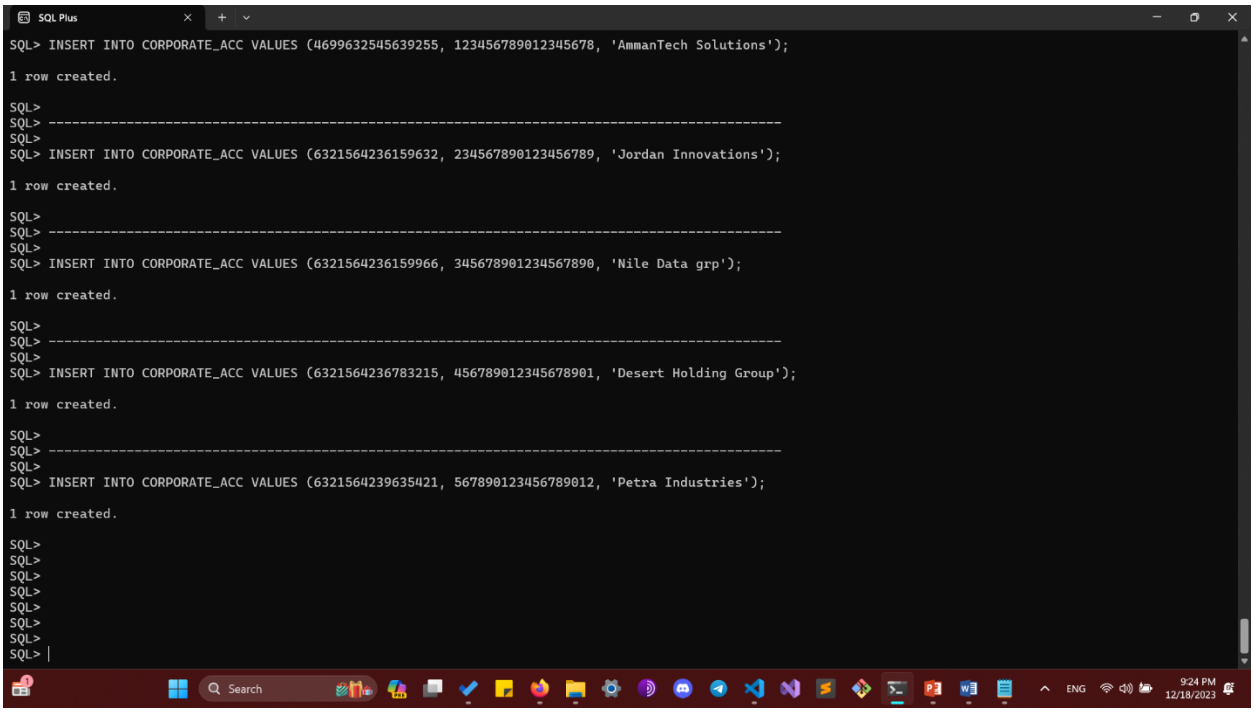
INSERT INTO CORPORATE_ACC VALUES (4699632545639255, 123456789012345678, 'AmmanTech Solutions');

INSERT INTO CORPORATE_ACC VALUES (6321564236159632, 234567890123456789, 'Jordan Innovations');

INSERT INTO CORPORATE_ACC VALUES (6321564236159966, 345678901234567890, 'Nile Data grp');

INSERT INTO CORPORATE_ACC VALUES (6321564236783215, 456789012345678901, 'Desert Holding Group');

INSERT INTO CORPORATE_ACC VALUES (6321564239635421, 567890123456789012, 'Petra Industries');



CREATE TABLE SAVING_ACC (ACCNUM NUMBER(16) CONSTRAINT S_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT S_ACCOUNTNUM_PK PRIMARY KEY , CARDNUM NUMBER(16) CONSTRAINT CARDNUM_UQ UNIQUE);

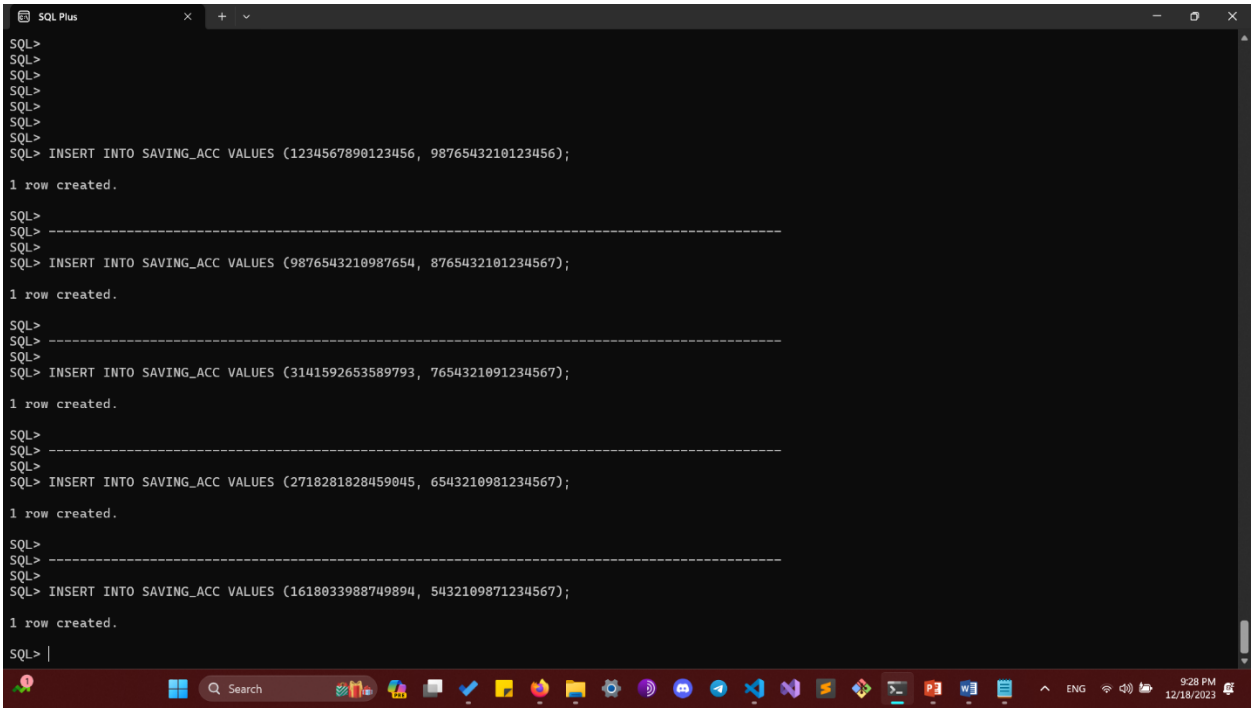
INSERT INTO SAVING_ACC VALUES (1234567890123456, 9876543210123456);

INSERT INTO SAVING_ACC VALUES (9876543210987654, 8765432101234567);

INSERT INTO SAVING_ACC VALUES (3141592653589793, 7654321091234567);

INSERT INTO SAVING_ACC VALUES (2718281828459045, 6543210981234567);

INSERT INTO SAVING_ACC VALUES (1618033988749894, 5432109871234567);



CREATE TABLE INV_DEP_ACC (ACCNUM NUMBER(16) CONSTRAINT IND_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT IND_ACCOUNTNUM_PK PRIMARY KEY , DPERIOD NUMBER(3) CONSTRAINT DPERIOD_NN NOT NULL);

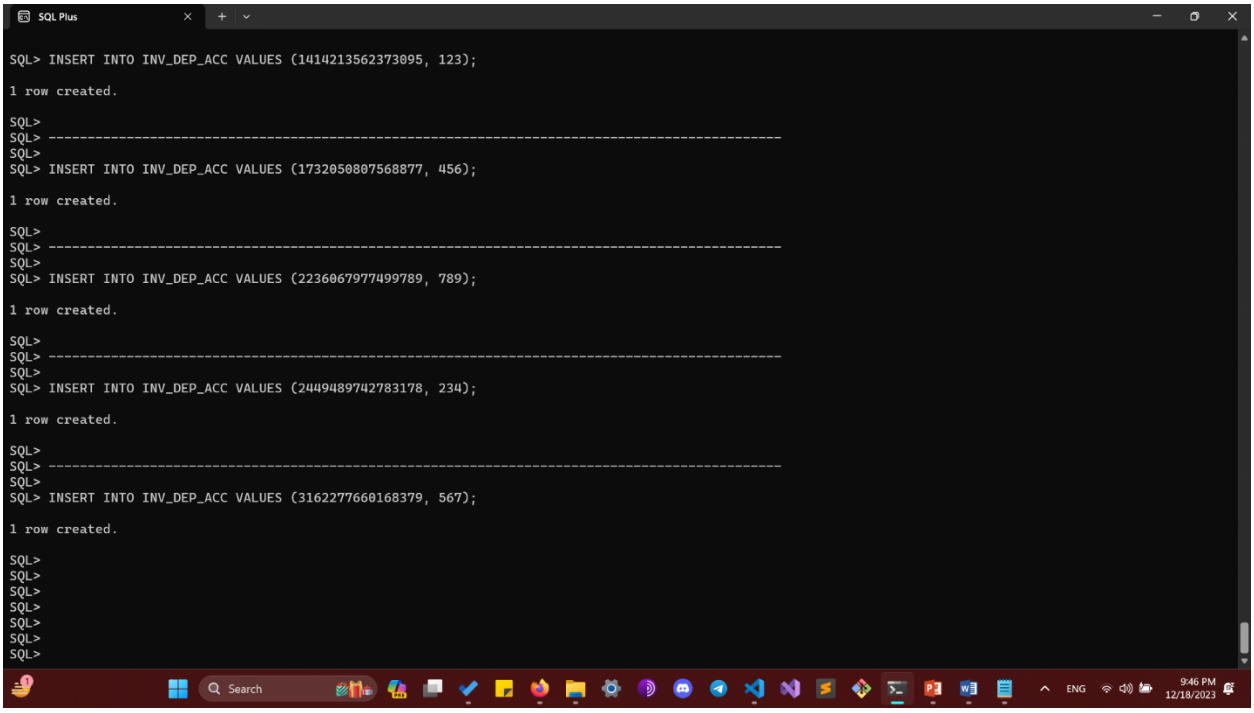
INSERT INTO INV_DEP_ACC VALUES (1414213562373095, 123);

INSERT INTO INV_DEP_ACC VALUES (1732050807568877, 456);

INSERT INTO INV_DEP_ACC VALUES (2236067977499789, 789);

INSERT INTO INV_DEP_ACC VALUES (2449489742783178, 234);

INSERT INTO INV_DEP_ACC VALUES (3162277660168379, 567);



CREATE TABLE CURRENT_ACC (ACCNUM NUMBER(16) CONSTRAINT CR_ACCOUNTNUM_FK REFERENCES BANK_ACCOUNT(ACCOUNTNUM) ON DELETE CASCADE CONSTRAINT CR_ACCOUNTNUM_PK PRIMARY

KEY , DCARDNUM NUMBER(16) CONSTRAINT DCARDNUM_UQ UNIQUE, CCARDNUM NUMBER(16) CONSTRAINT CCARDNUM_UQ UNIQUE, CHECKBOOKNUM NUMBER(6));

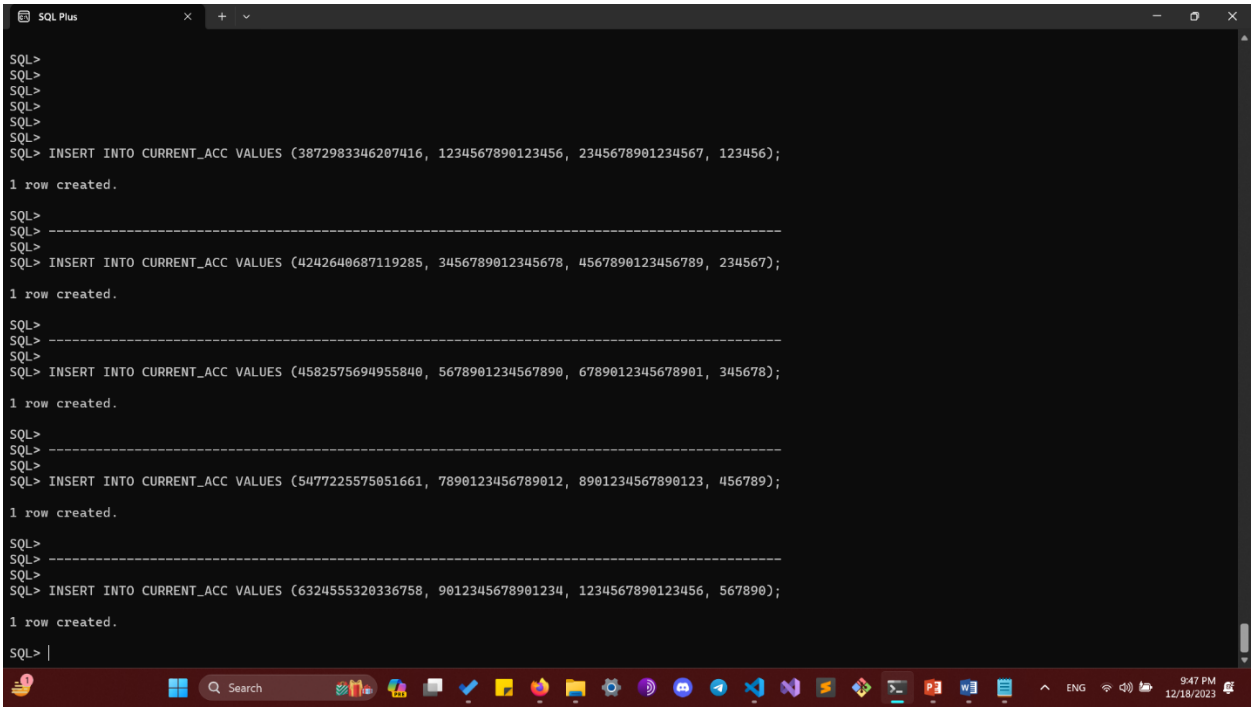
INSERT INTO CURRENT_ACC VALUES (3872983346207416, 1234567890123456, 2345678901234567, 123456);

INSERT INTO CURRENT_ACC VALUES (4242640687119285, 3456789012345678, 4567890123456789, 234567);

INSERT INTO CURRENT_ACC VALUES (4582575694955840, 5678901234567890, 6789012345678901, 345678);

INSERT INTO CURRENT_ACC VALUES (5477225575051661, 7890123456789012, 8901234567890123, 456789);

INSERT INTO CURRENT_ACC VALUES (6324555320336758, 9012345678901234, 1234567890123456, 567890);



CREATE TABLE SAVING_CUR (CUR VARCHAR2(3), ACCNUM NUMBER(16) CONSTRAINT SC_ACCNUM_FK REFERENCES SAVING_ACC(ACCNUM) ON DELETE CASCADE, CONSTRAINT SC_CUR_ACCNUM_PK PRIMARY KEY(CUR,ACCNUM));

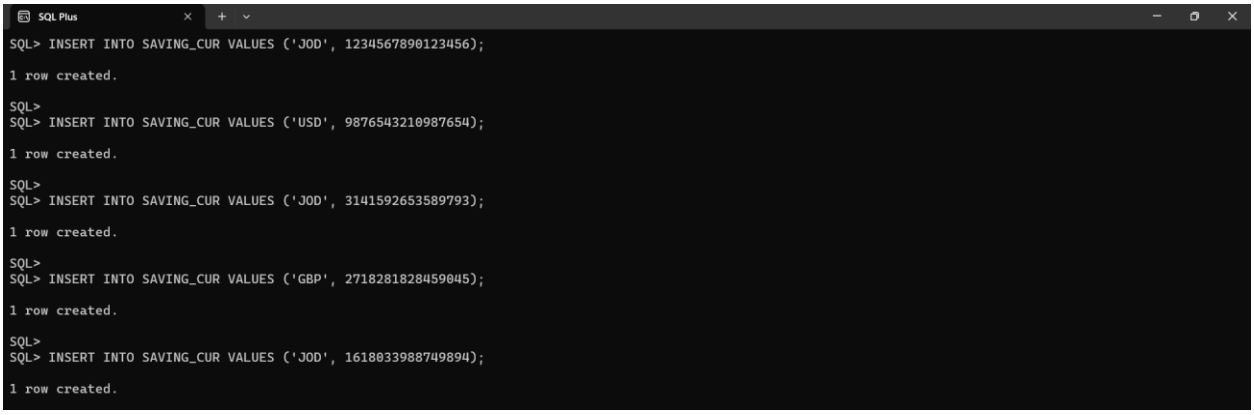
INSERT INTO SAVING_CUR VALUES ('JOD', 1234567890123456);

INSERT INTO SAVING_CUR VALUES ('USD', 9876543210987654);

INSERT INTO SAVING_CUR VALUES ('JOD', 3141592653589793);

INSERT INTO SAVING_CUR VALUES ('GBP', 2718281828459045);

INSERT INTO SAVING_CUR VALUES ('JOD', 1618033988749894);



CREATE TABLE INV_DEP_CUR (CUR VARCHAR2(3), ACCNUM NUMBER(16) CONSTRAINT IND_ACCNUM_FK REFERENCES INV_DEP_ACC(ACCNUM) ON DELETE CASCADE, CONSTRAINT INV_CUR_ACCNUM_PK PRIMARY KEY(CUR,ACCNUM));

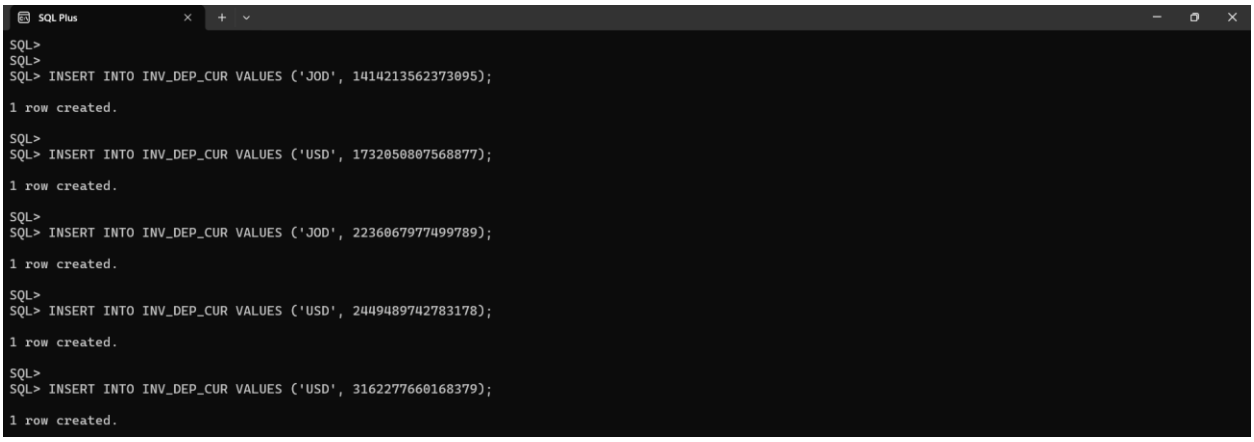
INSERT INTO INV_DEP_CUR VALUES ('JOD', 1414213562373095);

INSERT INTO INV_DEP_CUR VALUES ('USD', 1732050807568877);

INSERT INTO INV_DEP_CUR VALUES ('JOD', 2236067977499789);

INSERT INTO INV_DEP_CUR VALUES ('USD', 2449489742783178);

INSERT INTO INV_DEP_CUR VALUES ('USD', 3162277660168379);



CREATE TABLE CURRENT_CUR (CUR VARCHAR2(3), ACCNUM NUMBER(16) CONSTRAINT CC_ACCNUM_FK REFERENCES CURRENT_ACC(ACCNUM) ON DELETE CASCADE, CONSTRAINT CC_CUR_ACCNUM_PK PRIMARY KEY(CUR,ACCNUM));

INSERT INTO CURRENT_CUR VALUES ('EUR', 3872983346207416);

INSERT INTO CURRENT_CUR VALUES ('USD', 4242640687119285);

INSERT INTO CURRENT_CUR VALUES ('JOD', 4582575694955840);

INSERT INTO CURRENT_CUR VALUES ('JOD', 5477225575051661);

INSERT INTO CURRENT_CUR VALUES ('GBP', 6324555320336758);

```
SQL>
SQL> INSERT INTO CURRENT_CUR VALUES ('EUR', 3872983346207416);

1 row created.

SQL>
SQL> INSERT INTO CURRENT_CUR VALUES ('USD', 4242640687119285);

1 row created.

SQL>
SQL> INSERT INTO CURRENT_CUR VALUES ('JOD', 4582575694955840);

1 row created.

SQL>
SQL> INSERT INTO CURRENT_CUR VALUES ('JOD', 5477225575051661);

1 row created.

SQL>
SQL> INSERT INTO CURRENT_CUR VALUES ('GBP', 6324555320336758);

1 row created.
```

UPDATE CUSTOMER SET PHONENUM=0775598504 WHERE FNAME = 'Saed';

DELETE FROM WITHDRAW_DEPOSIT WHERE ACCNUM = 4242640687119285;

SELECT CUR, SAVING_CUR.ACCNUM FROM SAVING_CUR INNER JOIN SAVING_ACC ON SAVING_CUR.ACCNUM = SAVING_ACC.ACCNUM;

SELECT ACCNUM, COUNT(ACCNUM) AS “numofacc”, max(BALANCE) as “maxBalance”

from CURRENT_ACC ca, BANK_ACCOUNT ba

where ACCOUNTNUM = ACCNUM

group by ACCNUM;

SELECT FNAME, LNAME, COUNT(O.ACCNUM) AS NUM_ACC, SUM(BALANCE) AS TOT_BALANCE FROM CUSTOMER C, OWNS O, BANK_ACCOUNT B WHERE C.PDNUM = O.PDNUM AND O.ACCNUM = B.ACCOUNTNUM GROUP BY FNAME,LNAME;

```
SQL Plus

SQL> SELECT CUR, SAVING_CUR.ACCNUM FROM SAVING_CUR INNER JOIN SAVING_ACC ON SAVING_CUR.ACCNUM = SAVING_ACC.ACCNUM;

CUR     ACCNUM
--- ----------
JOD 1.2346E+15
USD 9.8765E+15
JOD 3.1416E+15
GBP 2.7183E+15
JOD 1.6180E+15

SQL> SELECT FNAME, LNAME, COUNT(O.ACCNUM) AS NUM_ACC, SUM(BALANCE) AS TOT_BALANCE FROM CUSTOMER C, OWNS O, BANK_ACCOUNT B WHERE C.PDNUM = O.PDNUM AND O.ACCN
UM = B.ACCOUNTNUM GROUP BY FNAME,LNAME;

FNAME           LNAME                    NUM_ACC TOT_BALANCE
--------------- -------------------- ---------- -----------
Saed            AlMajaly                      4      210171
Anas            AlAmro                        4      225604
Saqer           AlAdwan                       4      225604
Mohammed        Ahmad                         4      225604
Saeed           Issa                          4      225604
```

```
SQL> select ACCNUM, count(ACCNUM) as "numofacc", max(BALANCE) as "maxBalance"
  2  from CURRENT_ACC ca, BANK_ACCOUNT ba
  3  where ACCOUNTNUM = ACCNUM
  4  group by ACCNUM;

    ACCNUM   numofacc maxBalance
---------- ---------- ----------
3.8730E+15          1     156789
4.2426E+15          1      98765
4.5826E+15          1     123457
5.4772E+15          1      34568
6.3246E+15          1      54321

SQL>
```

```
SQL>
SQL> SELECT * FROM CUSTOMER;

     PDNUM RESPROOFDOC   PHONENUM NATIONALITY          FNAME           MNAME           LNAME
---------- ---------- ---------- -------------------- --------------- --------------- --------------------
2.0004E+10  1.2001E+11  799569874 Jordanian            Saed            Loay            AlMajaly
3.6259E+10  1.6955E+11  776529504 Jordanian            Anas            Ahmad           AlAmro
9.6326E+10  3.6026E+11  773623954 Canadian             Saqer           Metleg          AlAdwan
9.6232E+10  1.3010E+11  786325412 Jordanian            Mohammed        Maher           Ahmad
9.4232E+10  6.3524E+11  796541274 Jordanian            Saeed           Imad            Issa

SQL> SELECT * FROM BANK_ACCOUNT;

ACCOUNTNUM    BALANCE
---------- ----------
4.6996E+15        200
6.3216E+15      53967
6.3216E+15     153326
6.3216E+15       2678
6.3216E+15      15633
1.2346E+15     567890
9.8765E+15     190123
3.1416E+15      65432
2.7183E+15     456789
1.6180E+15     189012
1.4142E+15      76543

ACCOUNTNUM    BALANCE
---------- ----------
1.7321E+15     345679
2.2361E+15      78901
2.4495E+15      87654
3.1623E+15     234568
3.8730E+15     156789
4.2426E+15      98765
4.5826E+15     123457
5.4772E+15      34568
6.3246E+15      54321

20 rows selected.
```

```
SQL> SELECT * FROM LOAN;

ACCOUNTNUM      DELAY  MONAMOUNT LOANAMOUNT  MAXAMOUNT B_ACCOUNTNUM
---------- ---------- ---------- ---------- ---------- ------------
3.8730E+15         15       5000      20000      30000   3.8730E+15
4.2426E+15          8       7000      15000      25000   4.2426E+15
4.5826E+15         12       6000      18000      28000   4.5826E+15
5.4772E+15          5       8000      25000      35000   5.4772E+15
6.3246E+15         10       9000      22000      32000   6.3246E+15

SQL> SELECT * FROM WITHDRAW_DEPOSIT;

    ACCNUM      PDNUM
---------- ---------
4.2426E+15 2.0004E+10
4.6996E+15 2.0004E+10
3.8730E+15 3.6259E+10
4.5826E+15 2.0004E+10
5.4772E+15 3.6259E+10
6.3216E+15 2.0004E+10

6 rows selected.
```

SQL Plus

```
SQL> SELECT * FROM OWNS;

    ACCNUM      PDNUM
---------- ----------
4.6996E+15 2.0004E+10
6.3216E+15 2.0004E+10
6.3216E+15 2.0004E+10
6.3216E+15 2.0004E+10
6.3216E+15 3.6259E+10
6.3216E+15 3.6259E+10
6.3216E+15 3.6259E+10
6.3216E+15 3.6259E+10
6.3216E+15 9.6326E+10
6.3216E+15 9.6326E+10
6.3216E+15 9.6326E+10

    ACCNUM      PDNUM
---------- ----------
6.3216E+15 9.6326E+10
6.3216E+15 9.6232E+10
6.3216E+15 9.6232E+10
6.3216E+15 9.6232E+10
6.3216E+15 9.6232E+10
6.3216E+15 9.4232E+10
6.3216E+15 9.4232E+10
6.3216E+15 9.4232E+10
6.3216E+15 9.4232E+10

20 rows selected.

SQL> SELECT * FROM CORPORATE_ACC;

    ACCNUM CORREGNUMBER CORNAME
---------- ----------- --------------------
4.6996E+15   1.2346E+17 AmmanTech Solutions
6.3216E+15   2.3457E+17 Jordan Innovations
6.3216E+15   3.4568E+17 Nile Data grp
6.3216E+15   4.5679E+17 Desert Holding Group
6.3216E+15   5.6789E+17 Petra Industries
```

SQL Plus

```
SQL> SELECT * FROM SAVING_ACC;

    ACCNUM    CARDNUM
---------- ----------
1.2346E+15 9.8765E+15
9.8765E+15 8.7654E+15
3.1416E+15 7.6543E+15
2.7183E+15 6.5432E+15
1.6180E+15 5.4321E+15

SQL> SELECT * FROM INV_DEP_ACC;

    ACCNUM    DPERIOD
---------- ----------
1.4142E+15        123
1.7321E+15        456
2.2361E+15        789
2.4495E+15        234
3.1623E+15        567

SQL> SELECT * FROM CURRENT_ACC;

    ACCNUM   DCARDNUM   CCARDNUM CHECKBOOKNUM
---------- ---------- ---------- ------------
3.8730E+15 1.2346E+15 2.3457E+15       123456
4.2426E+15 3.4568E+15 4.5679E+15       234567
4.5826E+15 5.6789E+15 6.7890E+15       345678
5.4772E+15 7.8901E+15 8.9012E+15       456789
6.3246E+15 9.0123E+15 1.2346E+15       567890

SQL> SELECT * FROM SAVING_CUR;

CUR     ACCNUM
--- ----------
JOD 1.2346E+15
USD 9.8765E+15
JOD 3.1416E+15
GBP 2.7183E+15
JOD 1.6180E+15
```

```
SQL> SELECT * FROM INV_DEP_CUR;

CUR     ACCNUM
--- ----------
JOD 1.4142E+15
USD 1.7321E+15
JOD 2.2361E+15
USD 2.4495E+15
USD 3.1623E+15

SQL> SELECT * FROM CURRENT_CUR;

CUR     ACCNUM
--- ----------
EUR 3.8730E+15
USD 4.2426E+15
JOD 4.5826E+15
JOD 5.4772E+15
GBP 6.3246E+15
```